

Chapter 4:

Classical Conditioning:

Basic Phenomenon & Various Complexities

Outline for today

- Review of concepts in classical conditioning
 - 2 applied examples
- Extinction
 - 1 applied example
 - Spontaneous recovery
 - Disinhibition
- Stimulus generalization and discrimination

Conditioned Response

- Defined as “the response, often similar to (but never identical to) the UR, that is elicited by the CS”
- Salivating in response to the sound of the metronome is now called a *conditioned response* (CR).

Schematic Diagram: During Conditioning

Metronome:	Food	→	<i>Salivation</i>
NS (or CS)	US		UR

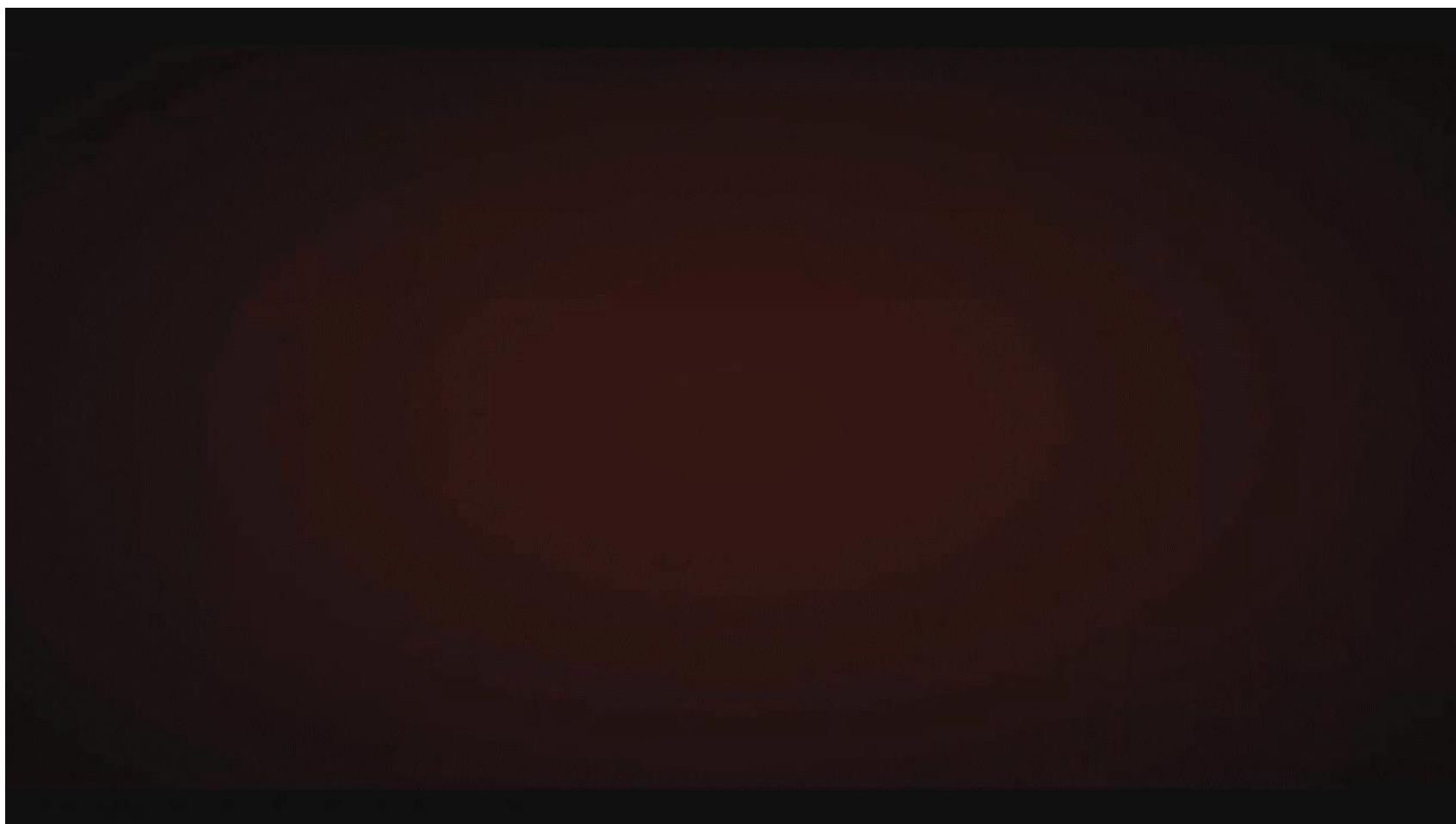
- Each pairing of the NS and US during conditioning is called a *conditioning trial*.
- Several conditioning trials are often needed
- During conditioning, the metronome can be labeled either an NS or a CS, because during this phase it begins as an NS and then becomes a CS.

Theme Songs

- Condition someone to theme song.

NS, CS, US, UR, CR



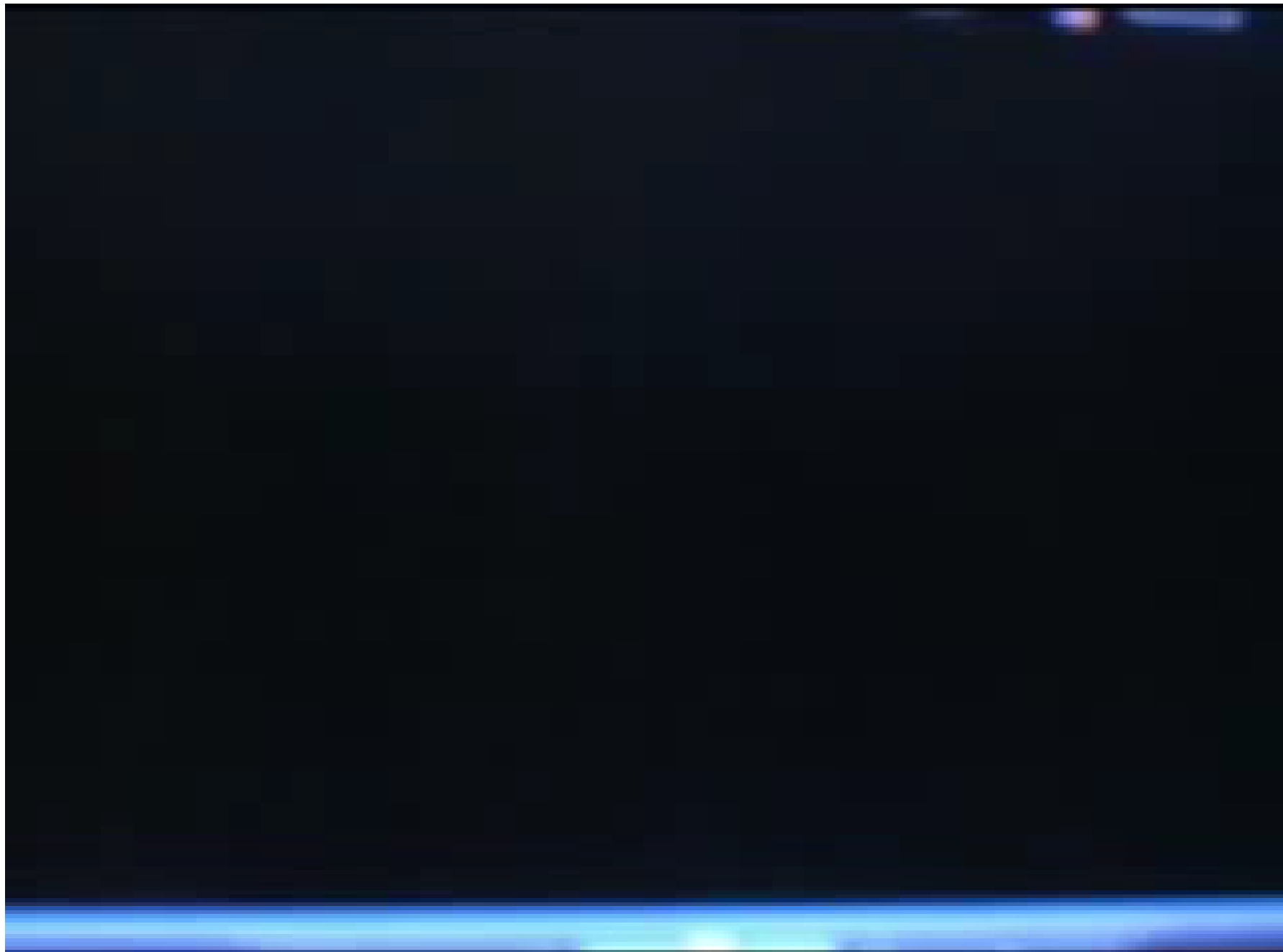


Theme Songs

- Condition someone to theme song.

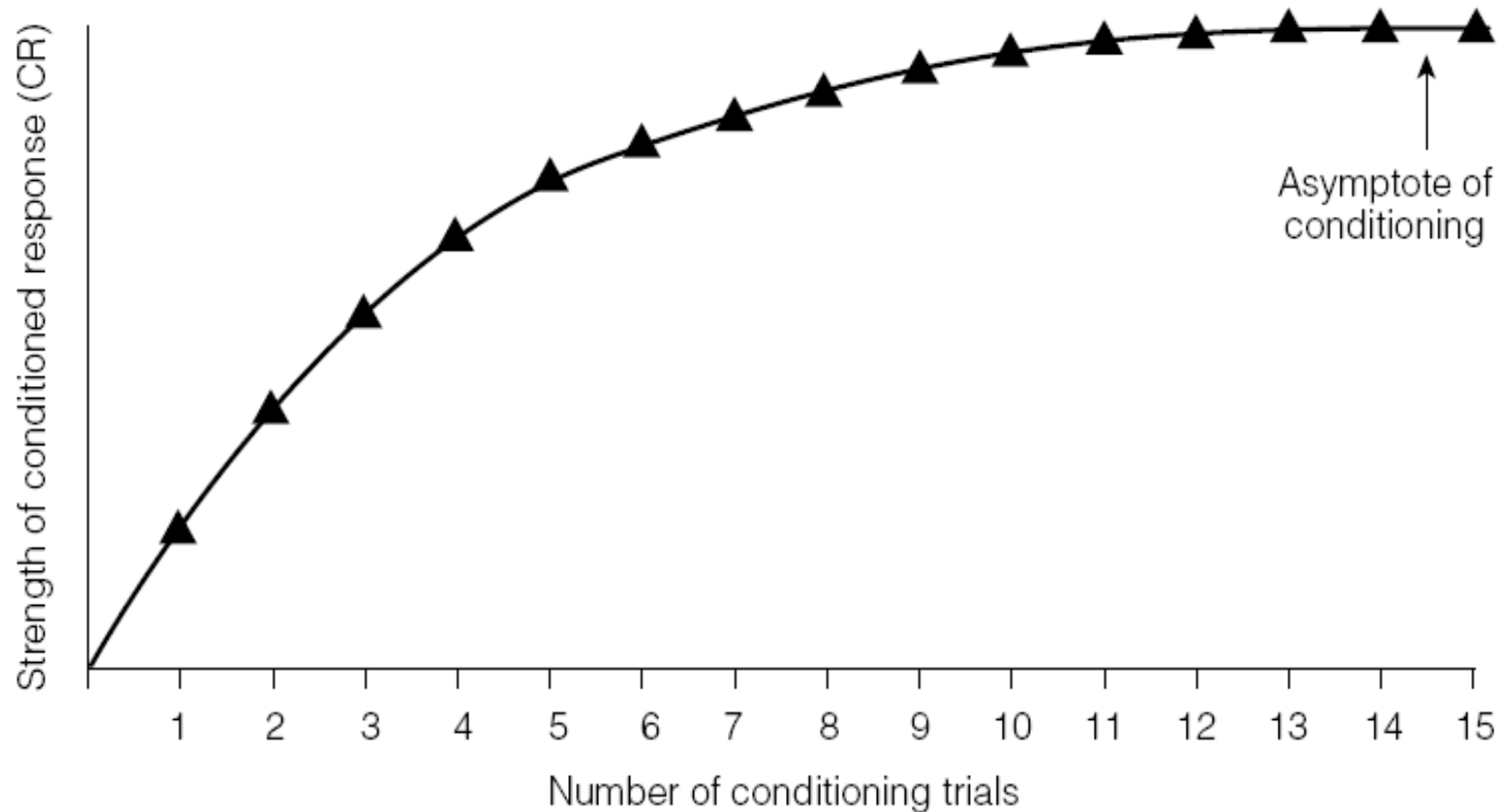
NS, CS, US, UR, CR



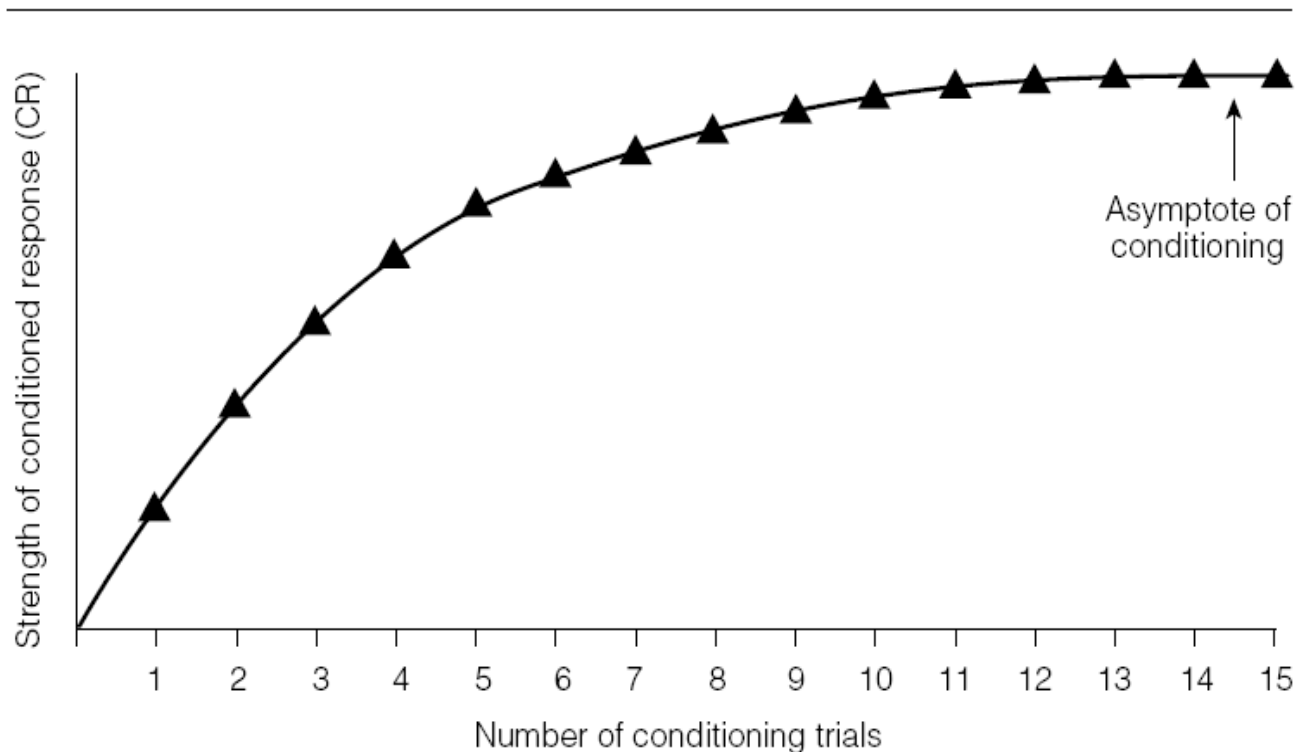


Typical Acquisition Curve

FIGURE 4.1 A typical acquisition curve in which strength of conditioning increases rapidly during the first few trials and then gradually levels off over subsequent trials.



Recitation: How can the unconditioned stimulus alter the acquisition curve?



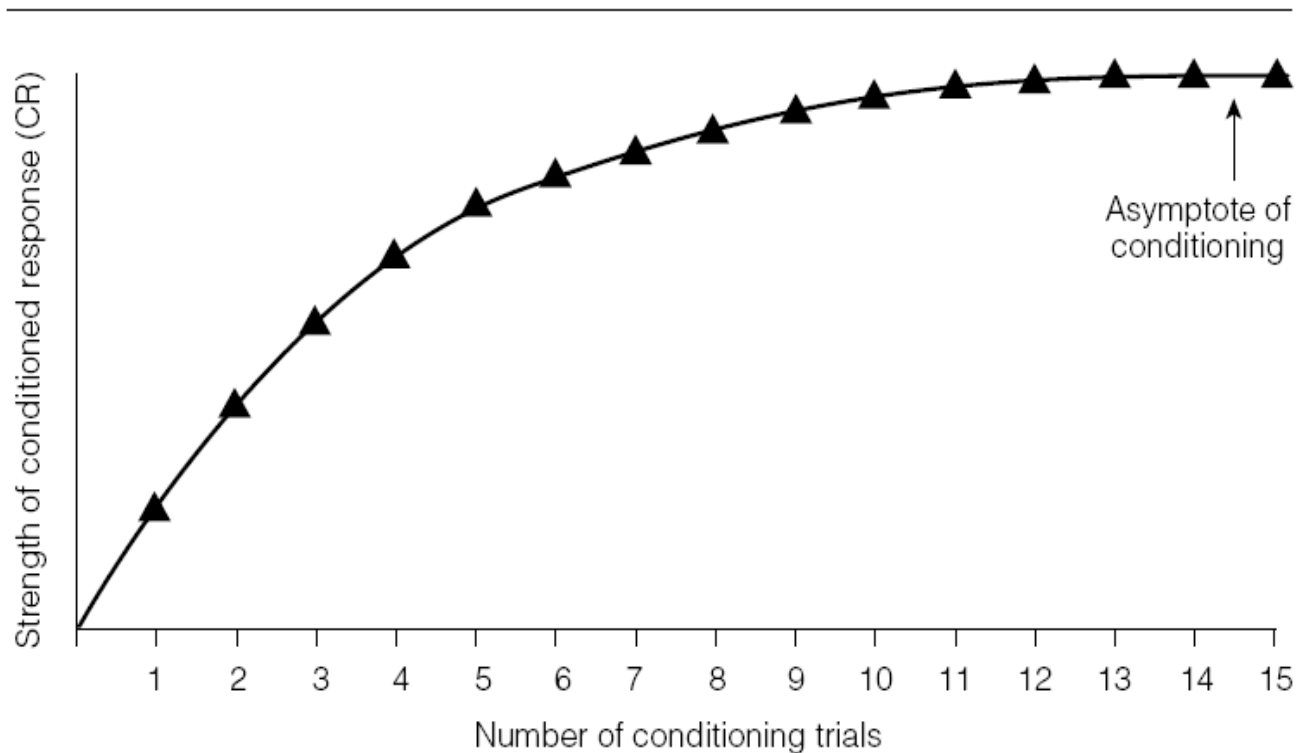
What influences the speed and asymptote of acquisition?

- Intensity of the US
- Example:
 - when the US consists of a large amount of food or a highly preferred food, conditioning is stronger.





Recitation: How can the neutral stimulus affect the acquisition curve?



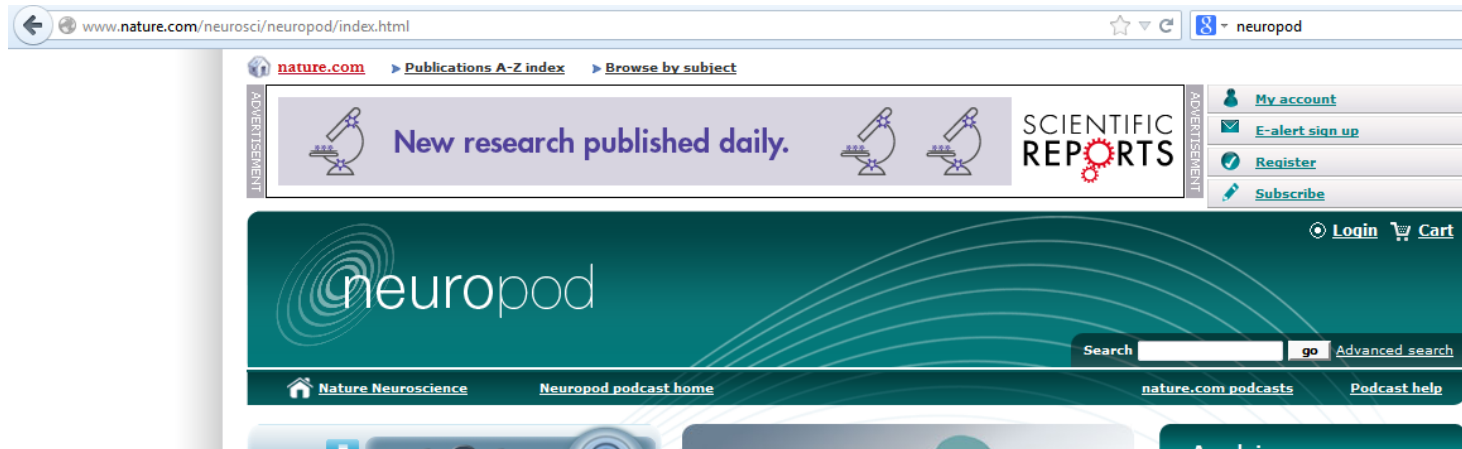
What influences the speed and asymptote of acquisition?

- Intensity of the neutral stimulus NS
- Example:
 - when the NS consists of a louder metronome, conditioning is stronger.



Are there limits to what we can learn?

- Can you learn in your sleep?
- Nature Neuropod
- August 2012 0:38 seconds



- Arzi A, Shedlesky L, Ben-Shaul M, Nasser K, Oksenberg A, Hairston IS, Sobel N. (August 2012) **Humans can learn new information during sleep.** Nat Neurosci. 2012 Aug 26.
- 0:35 August 2012 Neuropod

Sleep Experiment

- Unconditioned Stimulus
- Unconditioned Response
- Conditioned Stimulus
- Conditioned Response
- Conditioning Interval

Sleep Experiment

- Unconditioned Stimulus-
- 1 type of tone odor (unpleasant)
 - Rotten fish
- 2nd type of tone is other odor (pleasant)
 - Shampoo/perfume
- Unconditioned response-inhalation (respiration)
- Conditioned Stimulus-tone
- Conditioned Response-inhalation change
- Conditioning Interval-trace conditioning

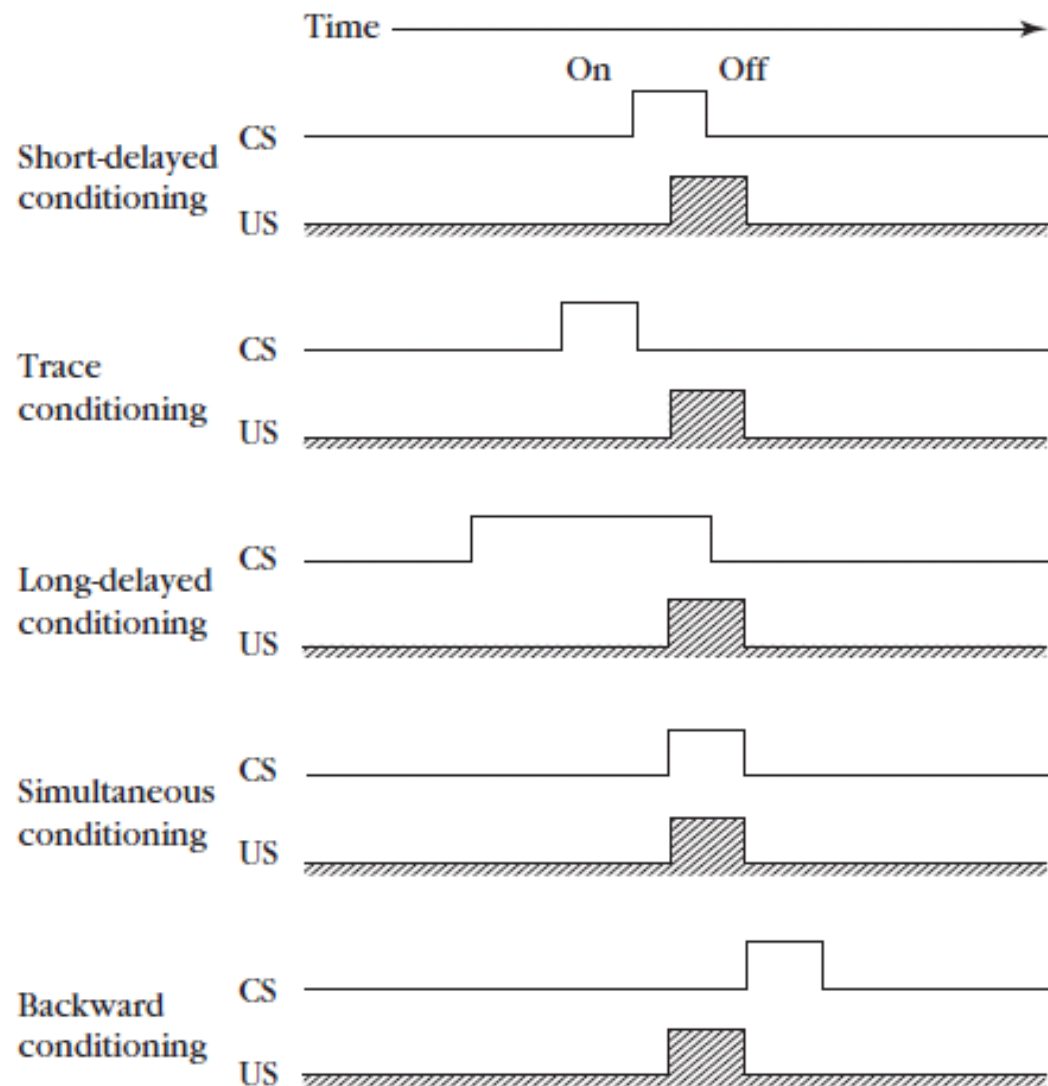


FIGURE 3.7

Five common classical conditioning procedures.

Review of Findings

- They repeatedly exposed the sleeping participants to pleasant odors, such as deodorant and shampoo, and unpleasant odors such as rotting fish and meat, and played a specific sound to accompany each scent.
- The subjects sniff strongly when they hear a tone associated with a pleasant smell, but only weakly in response to a tone associated with an unpleasant one.

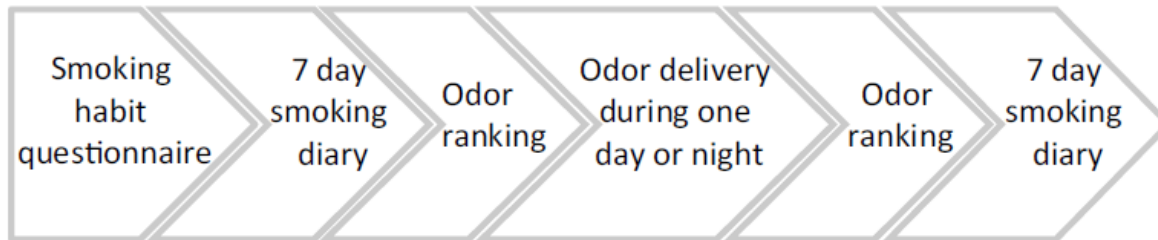
Behavioral/Cognitive

Olfactory Aversive Conditioning during Sleep Reduces Cigarette-Smoking Behavior

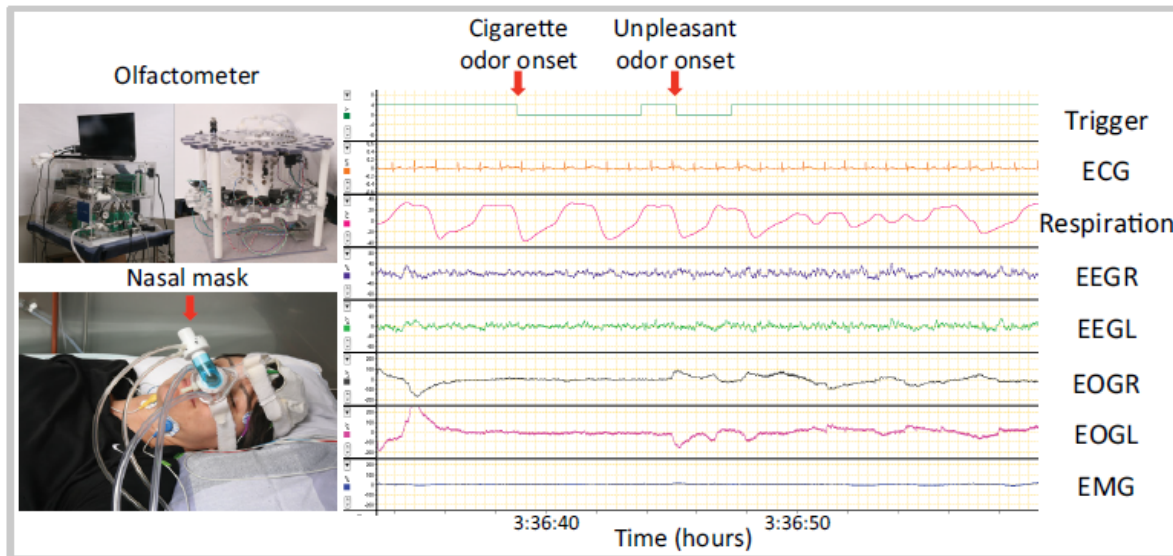
Anat Arzi, Yael Holtzman, Perry Samnon, Neetai Eshel, Edo Harel, and Noam Sobel

Department of Neurobiology, Weizmann Institute of Science, Rehovot, 7610001 Israel

A Experiment time line



B Polysomnography, respiration and odor delivery settings



i) Olfactory aversive conditioning:

X 10

X 10

X 10

ITl = 25-45 s

5 s 3 s 5 s 3 s 5 s

ii) Unpleasant odors:

X 10

X 10

X 10

ITl = 25-45 s

5 s 3 s 5 s 3 s 5 s

iii) Cigarette and unpleasant odors non-conditioned:

X 30

X 10

X 10

ITl = 25-45 s

5 s 3 s 3 s 5 s 5 s

X 10

X 10

X 10



X 10

X 10

X 10



X 30

X 10

X 10



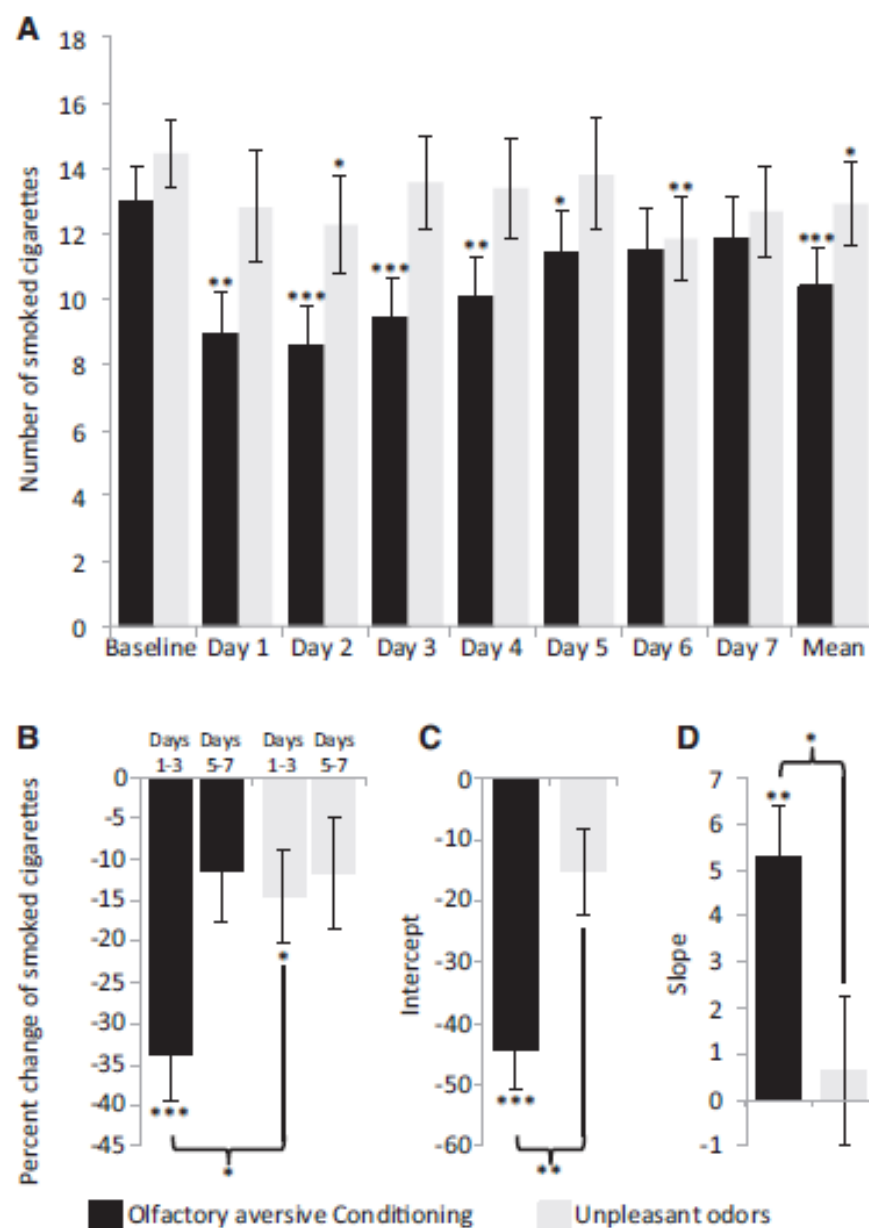


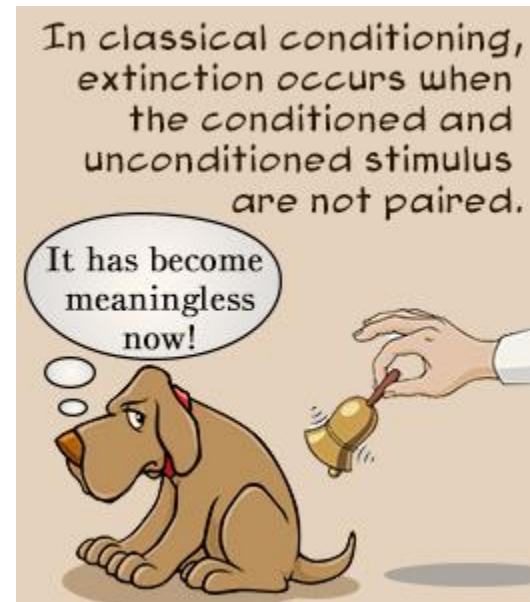
Figure 6. Olfactory aversive conditioning reduced smoking more effectively than unpleasant odors alone. **A**, Number of smoked

Recitation: Extinction

- What do you think of when you think of extinction in relationship to classical conditioning?

Extinction

- The Process
 - a conditioned response is weakened or eliminated when the CS is repeatedly presented in the absence of the US.
- The Procedure
 - the repeated presentation of the CS in the absence of the US.
- Example:
 - Procedure - present the metronome by itself
 - Process - salivation will eventually die out
 - Is this forgetting???



Notation for Extinction

Highway segment: Odor → *Lower Breathing*
NS US UR

Highway segment → *Lower Breathing*
CS CR

Highway segment → No change in breathing
“NS” —

What are some examples when you noticed you extinguished a certain behavior?

What happens to the CR?

- It has decreased in strength.
- It has not been completely eliminated.
- It can be reacquired quite rapidly when the CS (or NS) is again paired with the US.
- Example:
 - pair the metronome with food following an extinction procedure

Spontaneous Recovery

- The reappearance of a CS following a rest period after extinction.
- An extinguished response can reappear even in the absence of further pairings between the CS and US.

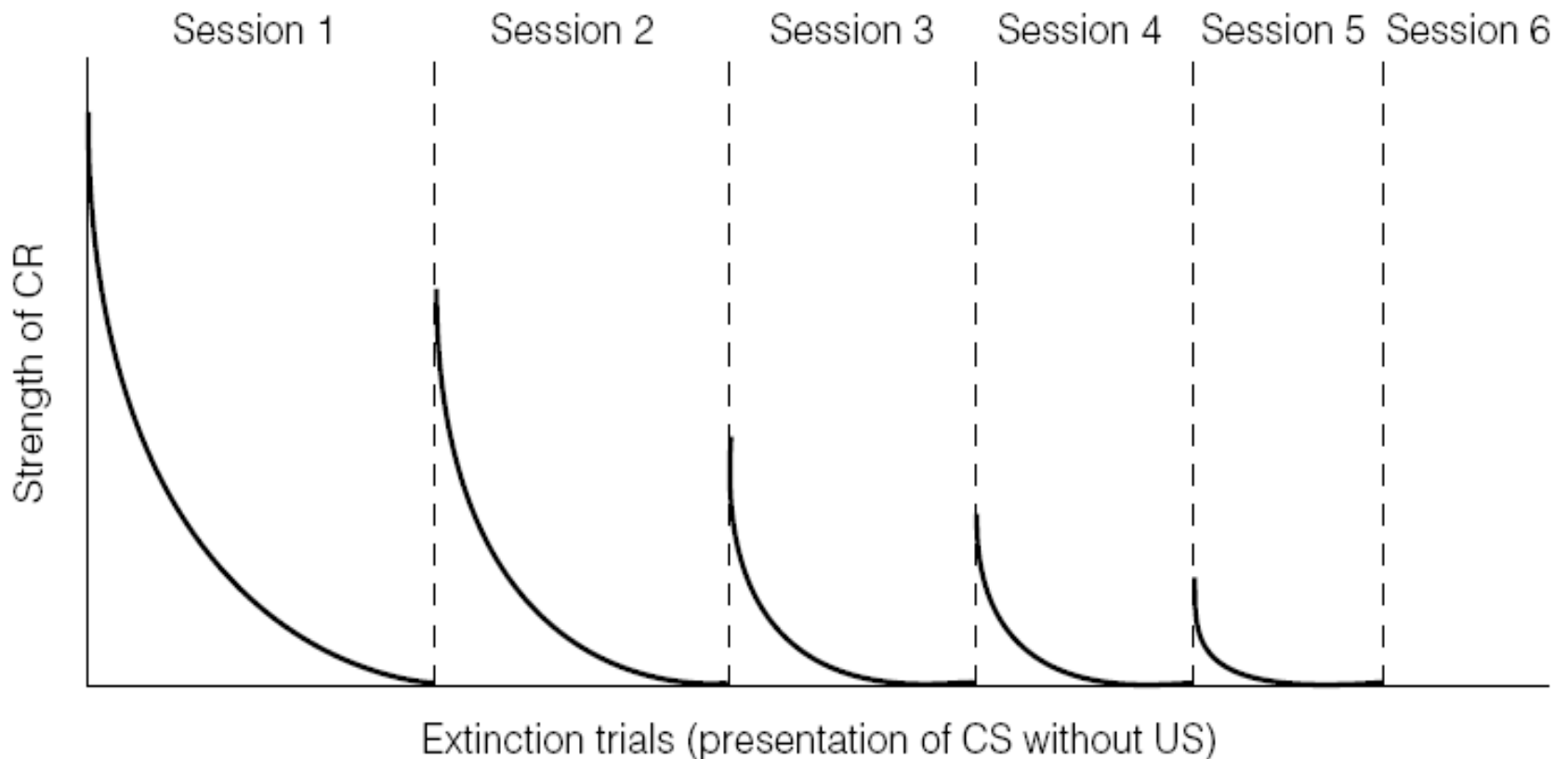


Spontaneous Recovery

- Each time the response recovers it is somewhat weaker and is extinguished more quickly than before.
- Example:
 - after several extinction sessions, the metronome will elicit little or no salivation.

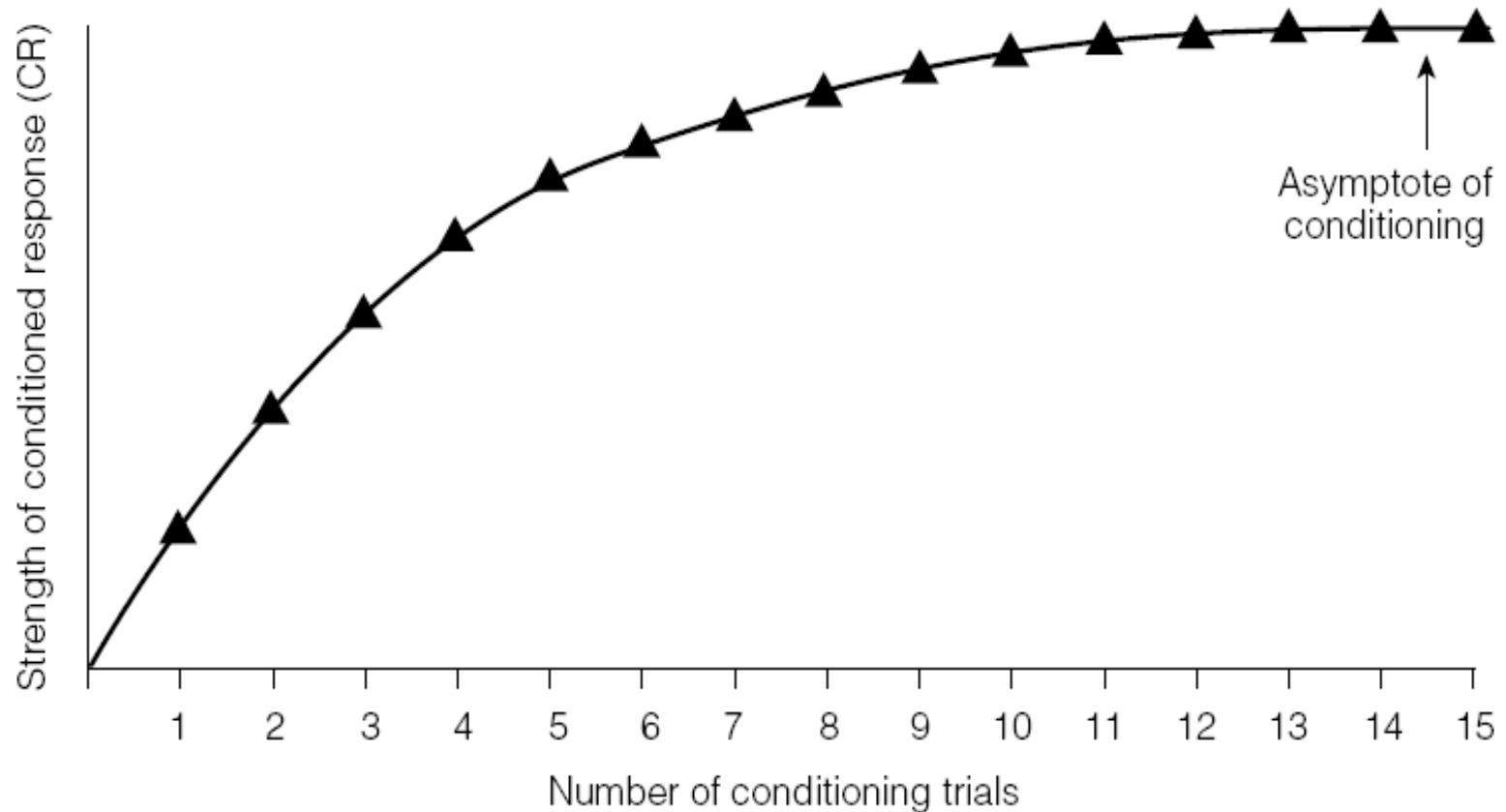
Spontaneous Recovery Graph

FIGURE 4.2 Hypothetical results illustrating a decline in spontaneous recovery across repeated sessions of extinction.



Typical Acquisition Curve

FIGURE 4.1 A typical acquisition curve in which strength of conditioning increases rapidly during the first few trials and then gradually levels off over subsequent trials.



- This is why you are nervous when confronted with your fear again

Learning Something New

- Extinction is not simply a process of unlearning the conditioning.
- Extinction involves learning something new.
- New conditioning inhibits the occurrence of the CR in the presence of the CS.
- Example:
 - The dog learns to inhibit the response of salivation to the metronome.



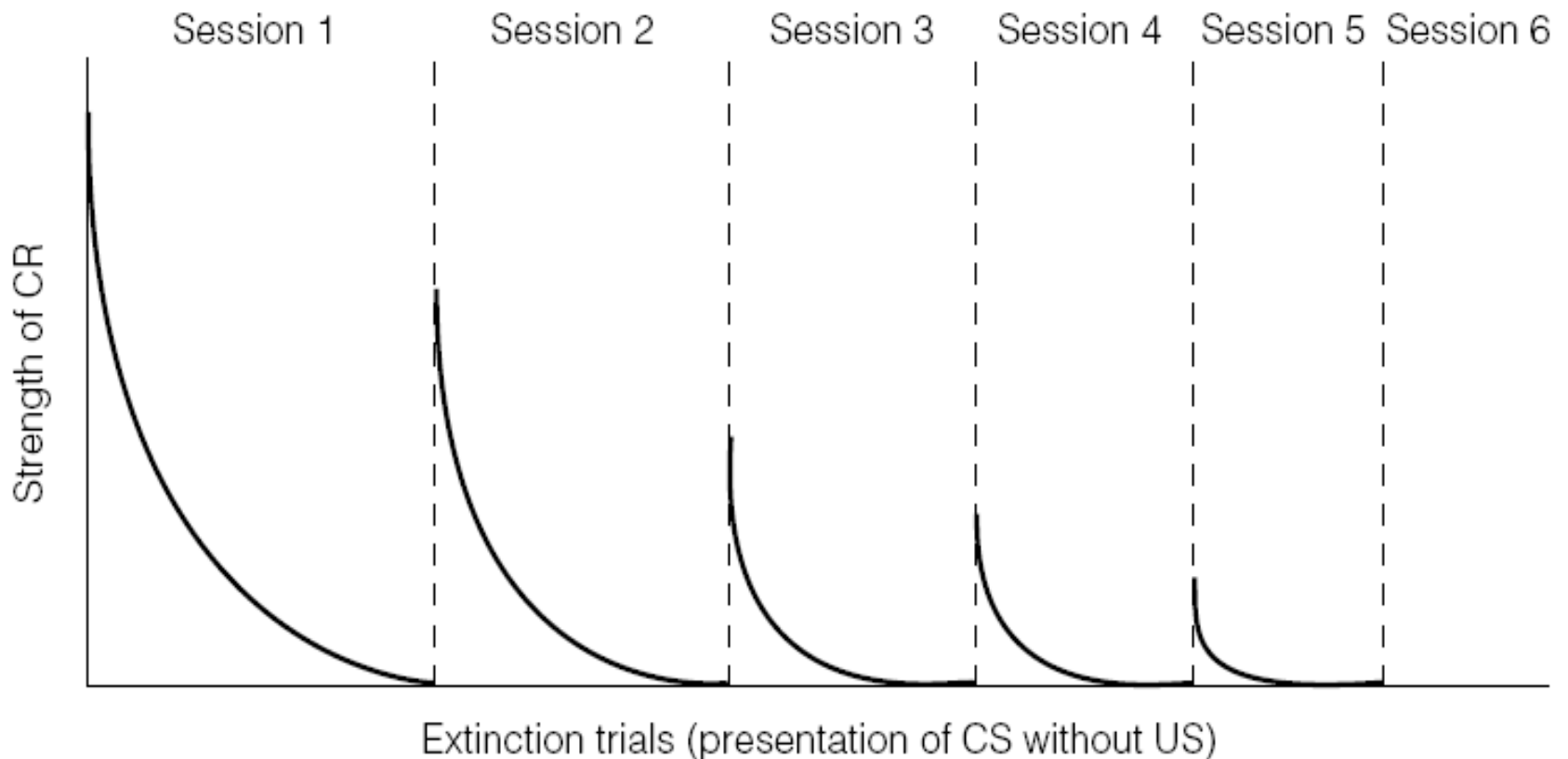
Disinhibition

- The sudden recovery of a response during an extinction procedure when a novel stimulus is introduced.
- Example:
 - If your anxiety while giving a speech gradually fades, it may suddenly recover when a noisy fan starts.



Spontaneous Recovery Graph

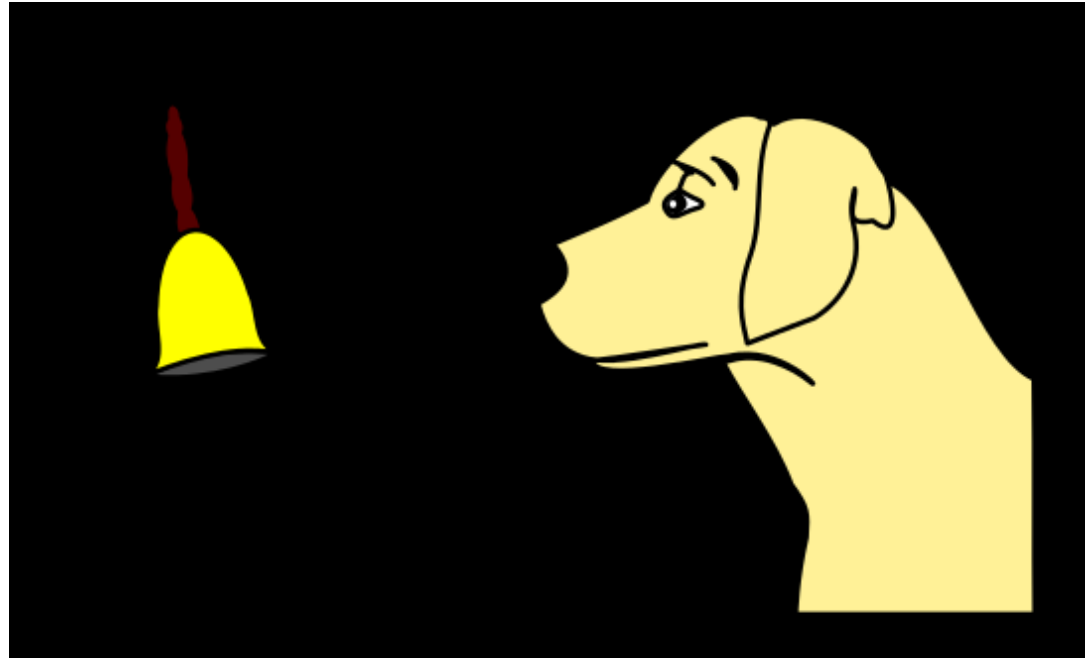
FIGURE 4.2 Hypothetical results illustrating a decline in spontaneous recovery across repeated sessions of extinction.



- This is why you want to give speeches in several different types of settings

Extinction Review

- Process of Extinction
- Spontaneous recovery
- Disinhibition



Stimulus Generalization

- The tendency for a CR to occur in the presence of a stimulus that is similar to the CS.
- The more similar the stimulus is to the original CS, the stronger the response.
- Example:



Stimulus Discrimination

- The tendency for a response to be elicited more by one stimulus than another.
- Example:
 - The dog salivates in the presence of the 2,000-Hz tone but not in the presence of a 1,900-Hz tone.
- This can be deliberately trained through *discrimination training*.

Discrimination Training

Conditioning Phase

2,000-Hz tone: Food → *Salivation*

NS US UR

1,900-Hz tone: No food → No Salivation

NS —

Test Phase

2,000-Hz tone → *Salivation*

CS+ CR

1,900-Hz tone → No salivation

CS- —

Discrimination

- The 2,000-Hz tone has become an excitatory CS (or CS+).
- It predicts the presentation of food.
- The 1,900-Hz tone has become an inhibitory CS (or CS–).
- It predicts the absence of food.



Discrimination learning and hobbies

- Wood working
- Cell phones
- Clothing





**AirPods
Pro**



Fake



Real

"A real pleasure... Blink brings wide-ranging insights
about our world and ourselves." —*Wall Street Journal*

#1 National Bestseller

WITH A NEW
AFTERWORD BY
THE AUTHOR

blink

by the author of Talking to Strangers



The Power of Thinking
Without Thinking

Malcolm Gladwell

What are some examples of stimulus discrimination?

- Ambulance
- Fire truck
- Police car
- Cell phone ring tones

Development of Disorders

- *Phobias* - overgeneralization of a fear response.
- Example:
 - A woman who overgeneralizes her fear of an abusive relationship may develop a fear of all relationships.
- *Experimental Neurosis* - neurotic-like symptoms which develop when exposed to extreme uncertainty.

Experimental Neurosis

Establishing an “*EXPERIMENTAL NEUROSIS*”

S+

S-



16:8



*Initial
Discrimination*



14:8



12:8



10:8



9:8



*Final
Discrimination*

Relationships and predictability

- Story at the beginning of the chapter
- A little uncertainty in a romantic relationship can be exciting, but extreme uncertainty might eventually become uncertain

End of Part 1 of Chapter 4

Associations and Birthday Parties



Higher-Order Conditioning

- stimulus that is associated with a CS can also become a CS.
- Example:
 - Wasp – Orange Tree

Wasp - Tree Example

Wasp: Sting $\rightarrow$ *Fear*

NS1 US UR

Wasp $\rightarrow$ *Fear*

CS1 CR

Orange Tree: Wasp $\rightarrow$ *Fear*

NS2 CS1 CR

Orange Tree $\rightarrow$ *Fear*

CS2 CR

A grandmother can be an example of high-order conditioning



DR. EGON SPENGLER
DR. RAY STANTZ
DR. ...

Shock Example: Recitation

 : →
NS1 US UR

 →
CS1 CR

 : →
NS2 CS1 CR

 →
CS2 CR

What are some other higher-order conditioning examples?







How Pennywise handles
Hurricane Irma

Stimulus Generalization

- The tendency for a CR to occur in the presence of a stimulus that is similar to the CS.



Third-Order Conditioning

- Pairing third stimulus with the CS.
- It is difficult to obtain.
- The conditioned response to a third-order conditioned stimulus (the CS3) is likely to be very weak.
- Example:
 - Advertising –

What happens when your celebrity does not behave?

- Michael Phelps
- Tiger Woods
- Lindsey Lohan
- Kobe Bryant
- Lance Armstrong
- What happens to the higher-order conditioning that you are working on

Sensory Preconditioning

- When one stimulus is conditioned as a CS, another stimulus it was previously associated with can also become a CS.



Wasp - Tree Example

Orange Tree: Wasp
NS2 NS1

Conditioning

Wasp: Sting → *Fear*
NS1 US UR

Wasp → *Fear*
CS1 CR

Orange Tree → *Fear*
CS2 CR

Latent Inhibition (CS preexposure effect)

- An unfamiliar stimulus is more readily conditioned than a familiar stimulus.
- Example:
 - If animal has metronome exposure 1st then conditioning, it will not condition
 - Sound of air conditioner turning on
- People with schizophrenia display less latent inhibition than is normal.

US Revaluation

- Postconditioning presentation of the US at a different level of intensity, thereby altering the strength of response to the previously conditioned CS.
- The value or magnitude of the US is changed.
- The intensity of response is dependent on the animal's most recent experience with the US.
- Example:
 - Giving the dog more food after conditioning it with less







Compound Stimulus

- Simultaneous presentation of two or more individual stimuli
- Example:
 - The sound of a metronome is presented at the same time as a light.
- Two Types:
 - Overshadowing
 - Blocking

Overshadowing

- the most salient member of a compound stimulus is more readily conditioned as a CS and thereby interferes with conditioning of the least salient member.
- Example:
 - a bright light and a faint-sounding metronome
 - Lead singer



BTS



Other Boy/Girl Bands

- NKOTB
- Big Time Rush
- The Jonas Brothers
- One Direction
- Backstreet Boys

Justice League







Notation for Overshadowing

[Bright light + Faint metronome]: Food $\rightarrow$ *Salivation*

NS

US

UR

[Bright light + Faint metronome] $\rightarrow$ *Salivation*

CS

CR

Bright light $\rightarrow$ *Salivation*

CS

CR

Faint metronome $\rightarrow$ No salivation

NS

—

Blocking

- the presence of an established CS interferes with conditioning of a **new** CS.
- consists of a neutral stimulus and a CS rather than two neutral stimuli (overshadowing) that differ in salience.

- Orange tree with wasp association
 - Whether the oranges are ripe or not would most likely be ignored by the presence of the orange tree
 - The additional stimulus needs to tell you something new or different

Notation for Blocking

[Light]: Food $\rightarrow$ *Salivation*

NS US UR

[Light] $\rightarrow$ *Salivation*

CS CR

Light + Metronome $\rightarrow$ *Salivation*

CS NS CR

Metronome $\rightarrow$ No salivation

NS —

Notation for Overshadowing

[Bright light + Faint metronome]: Food $\rightarrow$ *Salivation*

NS

US

UR

[Bright light + Faint metronome] $\rightarrow$ *Salivation*

CS

CR

Bright light $\rightarrow$ *Salivation*

CS

CR

Faint metronome $\rightarrow$ No salivation

NS

—



- Relate Pennywise to blocking and overshadowing.



Review of Concepts

- Other effectors
 - Sensory Preconditioning
 - Latent Inhibition
- Compound Stimulus
 - Overshadowing
 - Blocking

Temporal Conditioning

- a form of classical conditioning in which the CS is the passage of time.
- Example:
 - feeling hungry at noon

Occasion Setting

- a procedure in which a stimulus (i.e., an occasion setter) signals that a CS is likely to be followed by the US with which it is associated.
- Example:
 - Alcohol absent { Parents: Mild abuse → *Mild anxiety*
 - Alcohol present { Parents: Severe abuse → *Strong anxiety*

Any other examples?

- What signals do you send that you had a good/bad day?

Summary, continued

- Stimulus Generalization vs. Stimulus Discrimination
- Semantic Generalization
- Higher-order conditioning
- Sensory preconditioning
- Temporal conditioning
- Occasion setting

Summary, continued

- US revaluation
- Pseudoconditioning
- Processes that interfere with conditioning:
 - overshadowing
 - blocking
 - latent inhibition
 - external inhibition